NON-CONFIDENTIAL Executive Summary

Delhi's air quality routinely breaches the "Severe" threshold (AQI >400), posing a critical health risk to its 30 million residents. Current mitigation relies on the Graded Response Action Plan (GRAP), which is reactive, triggering blanket construction bans only after pollution peaks, causing massive economic disruption. Meanwhile, expensive hardware solutions like smog towers have proven ineffective beyond a 100-meter radius, and standard statistical models lack the precision to guide specific local actions.

Our project, "Urban Metabolism," overcomes these failures through a Physics-Informed AI model that forecasts pollution with high accuracy (RMSE 12.29 µg/m³). The core innovation is our ability to translate raw data into "Dynamic Zoning" policies. Unlike rigid city-wide bans, our system simulates "What-If" scenarios, demonstrating, for instance, that targeted traffic reduction can lower pollution by 17%.

Crucially, our solution integrates seamlessly into Delhi's existing governance architecture. We utilize live feeds from existing CPCB sensors and satellite data, requiring no new hardware. Our "Decision Support System" generates automated, actionable advisories aligned with DPCC protocols, allowing officials to preemptively divert traffic from low-wind corridors or halt construction only in specific "stagnation zones." By retrofitting intelligence onto legacy infrastructure, we transform reactive governance into precision urban planning, minimizing economic costs while safeguarding public health.

Project Title: Urban Metabolism: An AI-Driven Framework for Climate-Resilient City

Githubrepo: [Github link to see my model]

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Your Idea

We propose a Decision Support System (DSS) utilizing Physics-Informed AI to create a dynamic "Digital Twin" of Delhi's air quality. Unlike standard "black box" models, our architecture integrates atmospheric physics directly into the neural network, ensuring predictions are scientifically defensible for policy use.

Our one-year translational milestone is to transition our validated offline prototype into a live, cloud-based pilot for a high-risk zone, such as Anand Vihar. We aim to demonstrate "Dynamic Zoning" as a viable alternative to reactive governance. Instead of blunt,

city-wide bans that cripple the economy, our system enables hyper-local interventions, diverting traffic or halting construction only in specific micro-clusters where meteorological conditions indicate risk.

The impact is transformative. 30 million residents benefit from improved health through proactive warnings; municipal corporations gain evidence-based enforcement tools; and industries avoid the massive financial losses associated with indiscriminate blanket shutdowns. Commercially, this is positioned as a B2G SaaS platform for Smart Cities, with high scalability for other polluted urban centers globally.

Opportunity

Delhi routinely breaches "Severe" AQI thresholds, yet current solutions fail. The Graded Response Action Plan (GRAP) is reactive, triggering bans only *after* pollution peaks. High-cost hardware like Smog Towers has proven ineffective beyond a 100-meter radius. Structurally, "Urban Canyons" in dense zones like Laxmi Nagar trap pollutants, but planners lack tools to identify these specific bottlenecks.

Current methods are insufficient: statistical models lack accuracy, while CFD simulations are too slow for real-time use. Our approach offers the speed of AI with the reliability of physics. Crucially, the solution has profound Cross-Disciplinary Implications. It bridges the silos of Environment, Health, and Urban Planning like Environment to Health, Environment to Infrastructure, Environment to Water which will be discussed in depth later.

This synthesis creates a holistic "Urban Metabolism" framework, treating the city as a living system where infrastructure decisions ripple across health and environmental domains.

Preliminary Research or Rationale

Hypothesis & Technical Rationale: Decision-makers reject AI solutions when they function as opaque "black boxes" that cannot explain their reasoning. We hypothesize that for an AI to be adopted for critical urban planning, it must be constrained by physical reality. Our framework moves beyond pure pattern recognition by integrating the laws of wind dispersion directly into the model's core. If the model attempts to predict a scenario that defies physics, such as pollution rising during high winds—it is automatically penalized. This approach ensures the system generates outputs that are not only accurate but scientifically defensible, solving the critical "trust barrier" that prevents governments from adopting AI for public safety.

Foundational Evidence: Our preliminary prototype demonstrates high feasibility. Trained on meteorological and pollution data, the model achieved a Root Mean Squared Error (RMSE) of 12.29 µg/m³, a high-precision benchmark suitable for distinguishing between "Poor" and "Severe" air quality categories. Beyond forecasting, we validated the system's utility as a policy simulator through "What-If" scenarios. The model accurately predicted that a simulated 20% reduction in traffic emissions would yield a 17% drop in PM2.5, providing a quantitative "Return on Investment" metric for urban planners. Furthermore, our unsupervised clustering algorithm successfully generated an Urban Stress Index, automatically identifying "Stagnation Zones" and "Traffic Hotspots" to guide targeted infrastructure interventions.

Unique Insights & Team Advantage: Our approach is uniquely informed by global best practices in data utilization. During industry study treks to China, I observed how tech giants like Baidu and JD.com leverage pre-existing logistics and geospatial data to deliver fast, hyper-accurate services without the overhead of new infrastructure. Similarly, during my visit to Aereo in Bangalore, I witnessed how drone-based geospatial data is revolutionizing infrastructure and mining capabilities by providing high-resolution, actionable insights.

I aim to translate this "data-first" philosophy to the environmental sector. Instead of proposing expensive new hardware, we treat the city's existing sensor and satellite networks as a high-value asset, structuring this fragmented data through Physics-Informed AI to unlock actionable intelligence.Being a student of CS and AI with a minor in Economics and Public Policy and having previously worked at the intersection of technology and social causes, I am positioned to bridge the gap between complex algorithms and on-ground implementation. This project is not merely a technical exercise; it is an application of proven industrial efficiency models to solve a critical public health bottleneck, ensuring the team has the vision and practical context to succeed.

References

1. Raissi, M., et al. (2019). *Physics-informed neural networks*. Journal of Computational Physics.
2. Central Pollution Control Board (CPCB). (2023). *National Air Quality Index Manual*.
3. European Space Agency. *Sentinel-5P TROPOMI Level 2 Data Products*.

Project Plan and Continued Development

Goal The primary goal is to transition our validated prototype into a fully operational, cloud-based "Decision Support System" (DSS) for a pilot zone in Delhi (e.g., Anand Vihar). We aim to demonstrate that dynamic, data-driven zoning can reduce pollution exposure more efficiently than static bans.

Timeline (12 Months)

- ***Q1: Data Bridge & Infrastructure (Months 1-3):*** We will secure API access from the Central Pollution Control Board (CPCB) and the India Meteorological Department (IMD) to ingest real-time data streams. We will establish the cloud architecture necessary for high-frequency data processing.
- ***Q2: Model Refinement & Index Development (Months 4-6):*** The model will be retrained on live feeds to improve temporal accuracy. We will finalize the "Urban Stress Index" map, generating a real-time visualization interface that categorizes zones into "Stagnation," "Traffic Hotspot," or "Safe" archetypes.
- ***Q3: Pilot Deployment (Months 7-9):*** We will launch the dashboard for municipal stakeholders in the pilot zone. The system will begin generating "Weekly Urban Metabolism Advisories", automated reports recommending specific interventions (e.g., "Deploy anti-smog guns in Sector X due to predicted low wind").
- ***Q4: Validation & Feedback (Months 10-12):*** We will conduct a rigorous comparison of model recommendations against actual pollution outcomes. Using a "Human-in-the-Loop" feedback mechanism, we will refine the model based on enforcement realities and prepare the final impact report.

Phase II Pathway: Following the successful pilot, Phase II will focus on Scale-up and Commercialization. We aim to expand the DSS to cover the entire National Capital Region (NCR). The software will be packaged as a SaaS product for other Smart Cities in India, such as Mumbai and Kolkata, and aligned with the National Clean Air Programme (NCAP) objectives. This pathway positions the project for startup formation or integration into larger government procurement contracts.

Why Foundation Support? This project sits at the intersection of technology and public policy, a space difficult to navigate without institutional backing. We specifically need the Foundation’s support for cloud computing infrastructure to handle real-time geospatial data. Equally critical is the mentorship. While we have the technical capability, the Foundation’s network is vital to help us navigate the complex policy landscape and interface effectively with government bodies, ensuring our technical solution translates into real-world administrative action.

Intellectual Property, Scale or Policy Considerations

Intellectual Property & Scalability: This project generates distinct Intellectual Property centered on the Physics-Informed Loss Function and the proprietary Urban Stress Index (USI) algorithm. While raw pollution data is public domain, the specific methodology of mathematically constraining neural networks to adhere to the Advection-Dispersion equation creates a unique, protectable software asset. This algorithm is the core of our competitive advantage. It allows the model to achieve high accuracy with fewer computational resources than brute-force deep learning, making it highly scalable. We intend to position this as a SaaS platform (Software as a Service). Once validated in Delhi, the model is location-agnostic; it can be retrained on local data to serve other Smart Cities like Mumbai, Kolkata, or Bengaluru, creating a replicable model for the Global South.

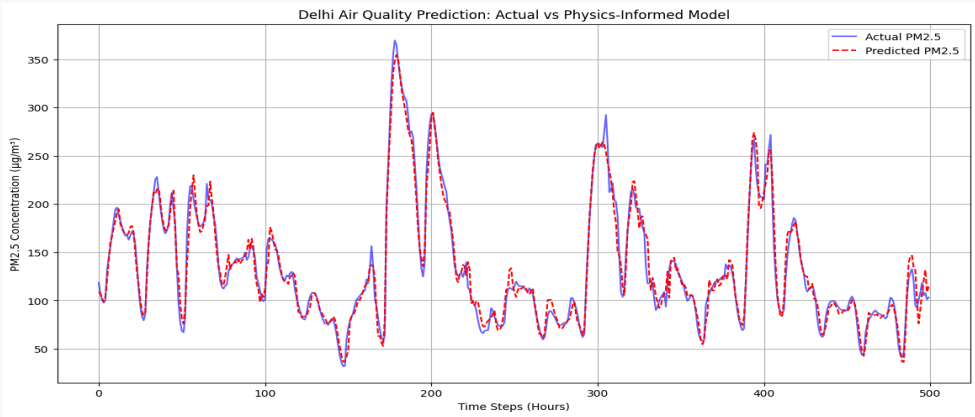
Collaboration & IP Ownership: We plan to collaborate with technical institutes like IIT Delhi for validation and with government bodies like CPCB for data access. This creates potential IP complexities. To mitigate risk, we will establish a clear "Research-to-Commercialization" agreement upfront. We propose a framework where the commercial IP rights (the algorithm and software code) remain vested with the project team to facilitate future startup formation. In contrast, the research partners receive credit for validation and joint publication rights. We seek the Foundation's legal mentorship to draft these MoUs professionally, ensuring that collaboration accelerates our timeline rather than entangling the project in ownership disputes.

Financial Interests & Conflicts: There are no existing financial interests, affiliations, or conflicts of interest to disclose. The project is currently unfunded and operates without ties to any existing startup or commercial organization. This clean slate is an advantage; it allows us to pursue the most effective solution without bias toward specific hardware vendors or legacy systems. Our sole interest is the successful translation of this technology into a deployable public good.

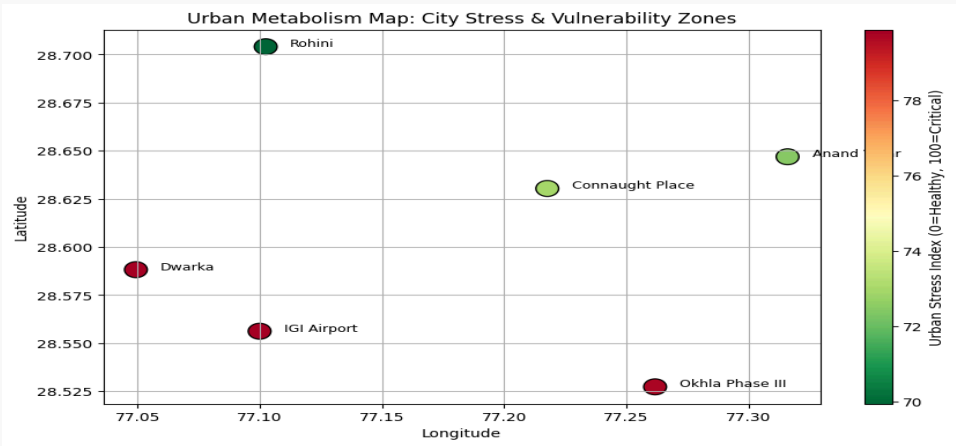
Policy & Systems Impact While the IP holds commercial value, the primary pathway for Year 1 is systems impact. We aim to fundamentally change how the Graded Response Action Plan (GRAP) is implemented. Currently, GRAP is a reactive, static protocol. By generating evidence that "Dynamic Zoning" (targeted interventions) reduces pollution more effectively than blanket bans, we will provide the quantitative basis for a policy amendment. We aim to introduce a "Predictive Zoning Clause" into Delhi's regulatory framework. This would allow authorities to enforce measures based on *forecasted* risk rather than current severity, preventing pollution peaks before they happen. This shift from reactive to proactive governance represents the ultimate institutional change we seek to drive.

Appendix Layout Plan

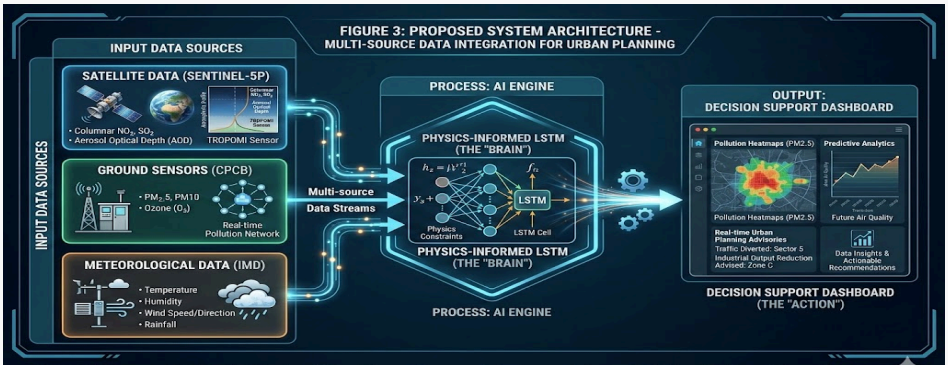
1. Figure 1: Model Validation (The "Accuracy" Proof)



2. Figure 2: Urban Stress Index Map (The "Planning" Proof)



3. Figure 3: System Architecture (The "How it Works" Diagram)



"Figure 3: Proposed System Architecture. Integrating multi-source data streams into a unified AI engine for real-time urban planning advisories." [cite: image_0.png]